A Productive Cultivation in Sea

Algae like *Kappaphycus alvarezii, Gracilaria edulis* and *Gelidiella acerosa* are commercially grown in the sea for harvesting the phycocolloids.



Sea Palm It is *Postelia palmaeformis* a brown alga.

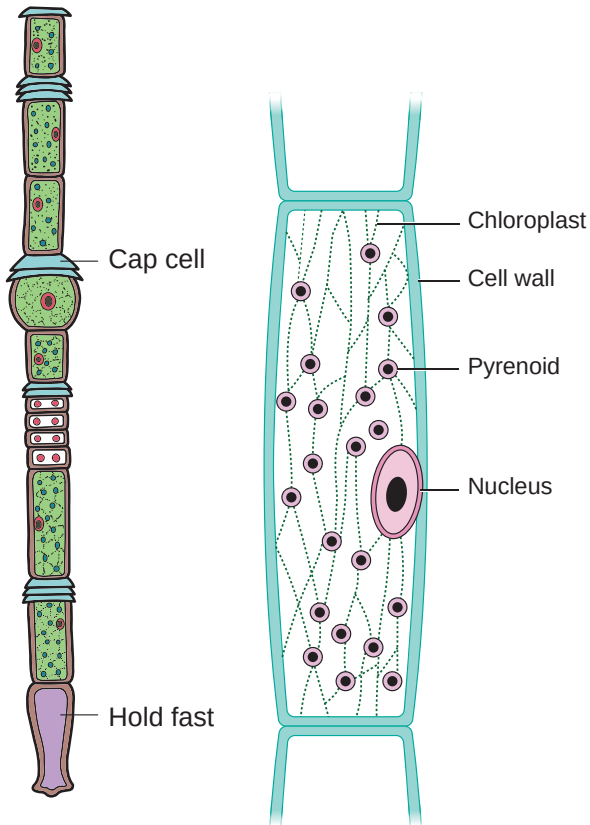
2.3.4 Oedogonium

Class – Chlorophyceae
Order - Oedogoniales
Family -Oedogoniaceae
Genus – *Oedogonium*

Oedogonium is a freshwater , filamentous alga and occurs in ponds, lakes and stagnant water. The filaments are attached to rocks. Some are epiphytic on aquatic plants. *Oedogonium terrestre* is a terrestrial form and grow in moist soils. The young filaments are attached but older ones are free floating.

Thallus structure

The thallus is filamentous ,multicellular and unbranched. All the cells of the filament are cylindrical except the basal and apical cell. The basal cell is colourless and forms hold fast. The proximal end of the hold fast extends to produce finger like projections which help the filament to attach on the substratum. The apical cell is rounded or elongated in shape. Each vegetative cell is cylindrical and possesses a thick cell wall. The inner layer is cellulosic and the outer layer is made up of pectin. A thin layer of chitin is present above the pectin layer. Next to the cell wall a plasma membrane is present. A large vacuole is present. The protoplasm contains reticulate chloroplast and it extends from one end of the cell to the other. A single nucleus and many pyrenoids are present. The distal end of some cells possess ring like markings called apical caps. Such cells are called cap cells. The presence of cap cell is characteristic feature of *Oedogonium* (Figure 2.5).



a) A single filament b) A cell enlarged

Figure 2.5: *Oedogonium*